MATH 345 Skeleton Notes
Unit 1: Vectors, matrices, and linear maps
Section 1.6: Exercises

Exercise tags. The tags below identify the primary outcomes and competency types for each exercise group. Individual problems may also involve earlier outcomes.

Matrices

Primary outcomes. U1-LO3. *Competencies.* P.

- Nicholson 2.1.3(f)
- Nicholson 2.1.8(b)
- Nicholson 2.1.15(b)

Matrix-vector multiplication

Primary outcomes. U1-LO4. *Competencies.* P+R.

- Nicholson 2.2.2(b)
- Nicholson 2.2.3(b)
- Nicholson 2.2.11(d)

Linear and affine maps

Primary outcomes. U1-LO5, U1-LO7. *Competencies.* P+C+R.

- Nicholson 2.6.2(b)
- Nicholson 2.6.13(b)

Matrix multiplication

Primary outcomes. U1-LO6. *Competencies.* P+R.

- Nicholson 2.3.1(b)
- Nicholson 2.3.3(b)
- Nicholson 2.3.5(b)
- Nicholson 2.3.16(b)

Additional applied and computational problems

These exercises connect the main Unit 1 ideas: vectors as data, cosine similarity, weighted averages, matrix-vector products, geometric matrix actions, composition, affine maps, and code interpretation. Solutions are collected in Appendix Unit 1 applied, geometric, and computational interpretation.

Exercise: Word-count vectors and cosine similarity

Use the dictionary `linear`, `matrix`, `data`. Let

$$\mathbf{q} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix},$$

$$\mathbf{D}_1 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \qquad \mathbf{D}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \mathbf{D}_3 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

1. Compute the cosine similarity between $\mathbf{q}$ and each document vector.

2. Rank the documents by cosine similarity.
3. Explain why $\mathbf{D}_1$ has cosine similarity 1 with $\mathbf{q}$.
4. Which document is orthogonal to $\mathbf{q}$?
5. What does that orthogonality mean in terms of the dictionary words?
6. Compute the Euclidean distance from $\mathbf{q}$ to each $\mathbf{D}_i$. Does the distance ranking match the cosine ranking?

Tags. `[U1-LO1, U1-LO2 | C+R | Core]`

Exercise: Weighted average of value vectors

Let

$$\mathbf{v}_1 = \begin{bmatrix} 10 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 10 \end{bmatrix}, \qquad \mathbf{v}_3 = \begin{bmatrix} 10 \\ 10 \end{bmatrix},$$

and let

$$\alpha = \begin{bmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{2} \end{bmatrix}.$$

1. Compute $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3$.
2. Explain why this is a weighted average.
3. Which value vector receives the largest weight?

Tags. `[U1-LO8 | P+C | Core]`

Exercise: Attention bridge: hand scores versus $K\mathbf{q}$

Use the token attention activity with tokens “small”, “red”, “bird”. Let

$$\mathbf{q} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

$$\mathbf{k}_{\text{small}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{k}_{\text{red}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathbf{k}_{\text{bird}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

1. Compute the three dot products

$$\mathbf{q} \cdot \mathbf{k}_{\text{small}}, \qquad \mathbf{q} \cdot \mathbf{k}_{\text{red}}, \qquad \mathbf{q} \cdot \mathbf{k}_{\text{bird}}.$$

2. Put the key vectors as rows of the matrix

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

   Compute $K\mathbf{q}$.

3. Explain why the entries of $K\mathbf{q}$ are the same scores from part (a).

4. Normalize the scores by dividing by their sum. What is the weight vector α?

Tags. `[U1-LO1, U1-LO4, U1-LO8 | C+R | Core]`

Exercise: Rows measure; columns contribute

Let

$$A = \begin{bmatrix} -1 & 4 & -5 \\ 3 & 1 & -2 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.$$

1. Compute $A\mathbf{x}$ using row dot products.

2. Compute $A\mathbf{x}$ as a linear combination of the columns of A.

3. In this example, what does the first row of A measure?
4. In this example, what do the coordinates $2, -3, 4$ tell you in the column-combination view?

Tags. `[U1-LO4 | P+R | Core]`

Exercise: The unit-square visualization

Let

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}.$$

1. Compute $A\mathbf{e}_1$ and $A\mathbf{e}_2$.
2. Compute $A\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.
3. Apply A to the four corners of the unit square:

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

4. Sketch the image of the unit square.
5. Which two transformed edges leaving the origin are the columns of A?

Tags. `[U1-LO5, U1-LO8 | C+R | Core]`

Exercise: Transformation matching

Match each matrix with the correct geometric description.

$$A_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad A_3 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix},$$

$$A_4 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad A_5 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \qquad A_6 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

1. Reflection across the y-axis.
2. Projection onto the x-axis.
3. Horizontal stretch.
4. Horizontal shear.
5. 90° counterclockwise rotation.
6. Reflection across the line $y = x$.

For each match, briefly justify your answer by describing what happens to $\mathbf{e}_1$ and $\mathbf{e}_2$.
Tags. `[U1-LO5 | C+R | Core]`

Exercise: Composition order: *RS* versus *SR*
Let

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \qquad R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

1. Compute RS.
2. Compute SR.
3. Are RS and SR equal?
4. Which product represents "apply S first, then apply R"?
5. Explain geometrically why the two orders produce different results.

Tags. `[U1-LO6 | P+R | Core]`

Exercise: Affine but not linear
Consider the transformation

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 1 \\ y \end{bmatrix}.$$

1. Compute $T(0,0)$.
2. Explain why T cannot be written in the form $T(\mathbf{x}) = A\mathbf{x}$ for any matrix A.
3. Write T in the affine form

$$T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}.$$

4. Is T linear, affine but not linear, or neither?

Tags. `[U1-LO7 | C+R | Core]`

Exercise: Code interpretation: sorting similarity scores

Consider the code:

```
import numpy as np

doc_names = np.array(["D1", "D2", "D3"])
scores = np.array([1.0, 0.5, 0.0])
ranking = doc_names[np.argsort(scores)[::-1]]
ranking
```

1. What is the value of `ranking`?
2. What does `np.argsort(scores)` return?
3. What does `[::-1]` do?
4. What is being ranked?
5. Is this ranking based on direction, distance, or alphabetical order?

Tags. `[U1-LO2, U1-LO8 | R+T | Core]`

Exercise: Transformer shape check

Suppose a sequence of token vectors is stored in a matrix

$$X \in \mathbb{R}^{6\times 5}.$$

Let

$$W_Q, W_K, W_V \in \mathbb{R}^{5\times 2},$$

and define

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V, \qquad S = QK^T.$$

A later weighting step produces an attention-weight matrix A_{att}, and the attention output is

$$H = A_{\mathrm{att}} V.$$

1. What are the shapes of Q, K, and V?
2. What is the shape of K^T?
3. What is the shape of $S = QK^T$?
4. What is the shape of A_{att}?
5. What is the shape of H?
6. What does the (i, j)-entry of S compute?
7. Which product forms the weighted averages of value vectors?
8. In this convention, are token vectors stored as rows or columns of X?

Tags. `[U1-LO3, U1-LO6, U1-LO8 | R+T | Core]`